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S2

Started on	Thursday, 17 September 2026, 9:00 AM
State	Finished
Completed on	Thursday, 17 September 2026, 9:10 AM
Time taken	10 mins 7 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than $\lfloor n / 2 \rfloor$ times. You may assume that the majority element always exists in the array.

Example 1:

Input: `nums = [3,2,3]`

Output: 3

Example 2:

Input: `nums = [2,2,1,1,1,2,2]`

Output: 2

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 104`
- `-231 <= nums[i] <= 231 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```

1 #include<stdio.h>
2 #include<stdlib.h>
3
4 int findMajorityElement(int nums[], int n) {
5     int candidate = nums[0];
6     int count = 1;
7
8     for (int i = 1; i < n; i++) {
9         if (count == 0) {
10             candidate = nums[i];
11             count = 1;
12         } else if (nums[i] == candidate) {
13             count++;
14         } else {
15             count--;
16         }
17     }
18     return candidate;
19 }
20
21 int main() {
22     int n;
23
24     if (scanf("%d", &n) != 1) {
25         return 0;
26     }
27
28     int *nums = (int*)malloc(n * sizeof(int));
29     if (nums == NULL) {
30         return 1;
31     }
32
33     for (int i = 0; i < n; i++) {
34         scanf("%d", &nums[i]);

```

```
35     }
36
37     int majority = findMajorityElement(nums, n);
38     printf("%d\n", majority);
39
40     free(nums);
41
42     return 0;
43 }
```

	Input	Expected	Got	
✓	3 3 2 3	3	3	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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